Numerical evaluation and matrix representation of the decay feed-down kernel

Notes accompanying `DecaySpectra.jl`

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These notes document (i) the numerical integration scheme used to evaluate the two- and three-body decay maps of the main notes (equation numbers below refer to *Maximum_Entropy_notes.pdf*) — the integration domain, its singularity structure, and the substitutions that make fixed-order Gauss–Legendre quadrature applicable — and (ii) how the maps are precomputed as matrices acting on a parent spectrum sampled on a fixed grid, including the block-Toeplitz (convolution) structure of the rapidity dependence.

1 The decay map is linear

After the analytic azimuthal integration, the daughter spectrum of the two-body decay $a \to b+c$ is [Eq. (32)]

$$\mathcal{F}_b(p_T,\eta_p,\phi_p) = \frac{B\,m_a}{4\pi\,p^a_{b|c}} \int_0^\infty q_T\,\mathrm{d}q_T \int_{-\infty}^{\infty} \mathrm{d}\eta_q\, \frac{\Theta\big(Q^2-D^2\big)}{\sqrt{Q^2-D^2}} \Big[\mathcal{F}_a(q_T,\eta_q,\phi_p+\alpha)+\mathcal{F}_a(q_T,\eta_q,\phi_p-\alpha)\Big], \tag{1}$$

with $M = m_{aT}m_{bT}$, $Q = q_T p_T$, $C = m_a E^a_{b|c}$, $D = M\cosh(\eta_q-\eta_p)-C$, $Z = D/Q$, $\alpha = \arccos Z$. The three-body decay [Eq. (59)] is the same kernel averaged over the effective mass μ of the $c+d$ pair with weight $p^a_b(\mu)\,p^{\tilde{c}}_c(\mu)/\mathcal{N}_{a\to bcd}$ [Eqs. (45)–(48)]. In Fourier modes, $\mathcal{F} = \sum_n \mathcal{F}_n e^{in\phi}$ [Eq. (62)], the two azimuthal roots produce the Chebyshev factor $2T_n(Z)$ [Eqs. (40), (41), (61)].

The essential observation is that (1) is *linear* in $\mathcal{F}_a$. Once the parent spectrum is represented by its values on a fixed grid, the map is a matrix, built once per decay and per mode; applying it to any source is a matrix–vector product.

2 Numerical evaluation of the kernel

Both decays reduce to the double integral

$$K(E_b) = \int q_T\,\mathrm{d}q_T \int \mathrm{d}\eta_q\, \frac{\Theta\big(Q^2-D^2\big)}{\sqrt{Q^2-D^2}}\,\big[\text{bracket}\big], \qquad p_b = \sqrt{E_b^2-m_b^2}, \quad C = m_a E_b, \tag{2}$$

where the bracket is the source sum of (1) (or $2T_n(Z)\mathcal{F}_{a,n}$ in mode form). The integrand is supported on a compact region with inverse-square-root boundary singularities; the strategy is to determine the support in closed form and remove every non-smooth feature by a substitution, so that plain tensor-product Gauss–Legendre quadrature converges rapidly.

2.1 The $\Delta\eta$ window

What the window is. Before the algebra, the physical content. The analytic azimuthal integration of the main notes rests on the on-shell δ-function $\delta(q \cdot p + m_a E_b)$ [Eq. (4)], which at fixed rapidities fixes the relative azimuth through $\cos\Delta\phi = Z$ [Eq. (19)], with

$$Z = \frac{M\cosh\Delta\eta - C}{Q} = \frac{D}{Q}. \tag{3}$$

A real opening angle $\Delta\phi$ exists *iff* $|Z| \le 1$ [Eq. (21)] — and $\Theta(Q^2 - D^2) = \Theta(1 - Z^2)$ is exactly the indicator of that condition. So the "$\Delta\eta$ window" answers a kinematic question: *for a daughter observed at (p_T, η_p), which parent momenta (q_T, η_q) can produce it?* The parent must sit at a rapidity separation $\Delta\eta$ for which some azimuthal orientation places the daughter at p_T; too small or too large a $\cosh\Delta\eta$ makes $|Z| > 1$ and no real $\Delta\phi$ exists. The endpoints $Z = \pm 1$ are the grazing configurations $\Delta\phi = 0, \pi$ (daughter emitted parallel or antiparallel to the parent's transverse momentum), and the $1/\sqrt{Q^2 - D^2} = 1/(Q\sqrt{1 - Z^2})$ factor is the Jacobian $|\mathrm{d}\cos\Delta\phi/\mathrm{d}\Delta\phi|^{-1}$ that diverges integrably there, as at any turning point.

The bounds. Throughout, $M = m_{aT} m_{bT} > 0$, $Q = q_T p_T \ge 0$ (with $Q > 0$ for interior nodes, $q_T, p_T > 0$), and $C = m_a E_b > 0$; all three depend on q_T and/or p_T but not on $\Delta\eta$. At fixed q_T the integrand is supported where the argument of the Θ function is positive:

$$Q^2 - D^2 \ge 0 \iff |D| \le Q \iff -Q \le M\cosh\Delta\eta - C \le Q. \tag{4}$$

Adding C and dividing by $M > 0$ (both operations preserve the inequalities) gives the two-sided condition

$$\frac{C - Q}{M} \le \cosh\Delta\eta \le \frac{C + Q}{M}. \tag{5}$$

This must be intersected with the range of the hyperbolic cosine, $\cosh\Delta\eta \ge 1$. Two of the three bounds compare trivially: $(C - Q)/M \le (C + Q)/M$ always (they differ by $2Q/M \ge 0$). Therefore

$$\cosh\Delta\eta \in \Big[\max\Big(1, \tfrac{C-Q}{M}\Big), \tfrac{C+Q}{M}\Big], \tag{6}$$

and this interval is nonempty *iff* its upper end clears the lower: since $\max(1, (C - Q)/M) \le (C + Q)/M$ reduces — whichever branch the max takes — to the single condition

$$\frac{C + Q}{M} \ge 1 \iff M \le C + Q, \tag{7}$$

because the branch $(C - Q)/M \le (C + Q)/M$ holds unconditionally. Condition (7) is the origin of the q_T band of Sec. 2.2.

To translate (6) into a set in $\Delta\eta$: on $\Delta\eta \ge 0$ the map $\Delta\eta \mapsto \cosh\Delta\eta$ is strictly increasing and continuous, so the preimage of the interval (6) is the interval $[\eta_{\mathrm{lo}}, \eta_{\mathrm{hi}}]$ with

$$\eta_{\mathrm{hi}} = \operatorname{arccosh}\frac{C + Q}{M}, \qquad \eta_{\mathrm{lo}} = \begin{cases} \operatorname{arccosh}\dfrac{C - Q}{M}, & C - Q > M, \\ 0, & C - Q \le M, \end{cases} \tag{8}$$

(the second case includes $C - Q \le 0$, where the lower constraint in (5) is vacuous since $\cosh \ge 1 > (C - Q)/M$). Since cosh is even, the full solution set is symmetric,

$$\big\{\Delta\eta : |\Delta\eta| \in [\eta_{\mathrm{lo}}, \eta_{\mathrm{hi}}]\big\} = \big[-\eta_{\mathrm{hi}}, -\eta_{\mathrm{lo}}\big] \cup \big[\eta_{\mathrm{lo}}, \eta_{\mathrm{hi}}\big] : \tag{9}$$

two disjoint intervals when $\eta_{\rm lo} > 0$ (the parent *must* be at different rapidity than the daughter), merging into the single interval $[-\eta_{\rm hi}, \eta_{\rm hi}]$ when $\eta_{\rm lo} = 0$. The change between the two topologies as q_T varies is the origin of the logarithmic singularity of Sec. 2.3.

Geometrically the two regimes read as follows. When $\eta_{\rm lo} > 0$ (i.e. $C - Q > M$, meaning C is large: the parent is slow or the daughter hard) the parent cannot sit at the same rapidity as the daughter — a co-moving parent would have to emit the daughter with $|Z| > 1$ — so the allowed set is two rapidity bands straddling η_p, symmetric about it. As q_T increases, Q grows, $C - Q$ eventually drops below M, the inner gap closes ($\eta_{\rm lo} \to 0$), and the bands merge: now a parent *at* the daughter's rapidity can feed it (with $\Delta\phi$ given by the interior value $Z(0) = (M - C)/Q \in (-1, 1)$). The merging point is exactly q_T^* of Sec. 2.3. The upper edge $\eta_{\rm hi}$ always exists (some maximal rapidity separation beyond which the boost carries the daughter past p_T for every azimuth) and is always a genuine $1/\sqrt{\ }$ singularity; the lower edge is singular only in the two-band regime. Both rapidity signs $\Delta\eta \gtrless 0$ contribute; they are evaluated separately because the source need not be symmetric in rapidity. At the window edges $Z = \pm 1$ and $Q^2 - D^2$ vanishes *linearly* in $\Delta\eta$: writing $Q^2 - D^2 = (Q - D)(Q + D)$ (the factored form is also what is computed — it avoids the cancellation of subtracting two large squares), one factor vanishes at each edge while $\mathrm{d}D/\mathrm{d}\Delta\eta = M \sinh\Delta\eta \neq 0$ there, so the zero is simple and the integrand diverges as an integrable $1/\sqrt{\ }$. When $\eta_{\rm lo} = 0$ (merged window) the lower edge is *not* singular — $Z(0) = (M - C)/Q$ lies inside $(-1, 1)$ — and the cosine map's clustering there is merely harmless. Two numerical guards accompany the window: q_T nodes with $(C + Q)/M \leq 1$ are skipped (roundoff can push a node marginally outside the support), and inside the loop $S_2 = (Q - D)(Q + D) \leq 0$ nodes are dropped for the same reason.

2.2 The q_T band

By (7), the window is nonempty exactly when

$$m_{aT}\, m_{bT} \;\leq\; C + q_T p_T. \tag{10}$$

Both sides of (10) are strictly positive ($C = m_a E_b > 0$ and $m_{aT} m_{bT} > 0$), so squaring preserves the inequality in both directions and is an equivalence. Squaring and expanding with $m_{aT}^2 = m_a^2 + q_T^2$, $m_{bT}^2 = m_b^2 + p_T^2$:

$$\begin{aligned} \big(m_a^2 + q_T^2\big)\big(m_b^2 + p_T^2\big) \;&\leq\; C^2 + 2C p_T\, q_T + q_T^2 p_T^2, \\ m_a^2 m_b^2 + m_a^2 p_T^2 + m_b^2 q_T^2 + q_T^2 p_T^2 \;&\leq\; C^2 + 2C p_T\, q_T + q_T^2 p_T^2. \end{aligned} \tag{11}$$

The $q_T^2 p_T^2$ terms cancel; using $m_a^2 m_b^2 + m_a^2 p_T^2 = m_a^2 m_{bT}^2$ and collecting powers of q_T,

$$m_b^2\, q_T^2 - 2C p_T\, q_T + m_a^2 m_{bT}^2 - C^2 \;\leq\; 0, \tag{12}$$

a upward parabola in q_T: the inequality holds between its roots. The quarter-discriminant is

$$\begin{aligned} \tfrac{1}{4}\Delta &= (C p_T)^2 - m_b^2\big(m_a^2 m_{bT}^2 - C^2\big) = C^2\big(p_T^2 + m_b^2\big) - m_a^2 m_b^2\, m_{bT}^2 \\ &= m_{bT}^2\big(C^2 - m_a^2 m_b^2\big) = m_{bT}^2\, m_a^2\big(E_b^2 - m_b^2\big) = m_{bT}^2\, m_a^2\, p_b^2 \;\geq\; 0, \end{aligned} \tag{13}$$

where the second line used $p_T^2 + m_b^2 = m_{bT}^2$, then $C = m_a E_b$, then $p_b^2 = E_b^2 - m_b^2$. The discriminant is a perfect square — the roots are always real, as they must be (the band exists for every open decay). The quadratic formula then gives

$$q_T^{\pm} = \frac{C p_T \pm m_a p_b\, m_{bT}}{m_b^2} = \frac{m_a\big(E_b\, p_T \pm p_b\, m_{bT}\big)}{m_b^2}, \qquad q_T \in \big[\max(0, q_T^-),\, q_T^+\big]. \tag{14}$$

The sign of the lower root follows from $\big(E_b p_T\big)^2 - \big(p_b m_{bT}\big)^2 = E_b^2 p_T^2 - p_b^2\big(m_b^2 + p_T^2\big) = p_T^2\big(E_b^2 - p_b^2\big) - p_b^2 m_b^2 = m_b^2\big(p_T^2 - p_b^2\big)$, so $q_T^- > 0 \iff p_T > p_b$. The lower root is positive iff $p_T > p_b$: a daughter harder than the rest-frame momentum can only come from a moving parent, while for $p_T < p_b$ even a parent at rest contributes and the band starts at $q_T = 0$.

The band width, $q_T^+ - q_T^- = 2m_a p_b m_{bT}/m_b^2$, carries the m_b^{-2} of the roots: light daughters have enormous bands (for $\rho \to \pi\pi$ at $p_T = 0.6\,\mathrm{GeV}$ the band is $[0.5,\, 18]\,\mathrm{GeV}$), which is why the feed-down matrices of Sec. 4 are effectively dense. In practice the reach in q_T is limited not by the band but by the source: for a thermal-like parent spectrum $\sim e^{-(m_{aT}-m_a)/T_{\mathrm{eff}}}$ the integrand is negligible beyond

$$q_T^{\max} \;=\; \sqrt{\big(m_a + T_{\mathrm{eff}}\ln(1/\varepsilon)\big)^2 - m_a^2}, \tag{15}$$

(ε the tolerated relative tail, e.g. 10^{-8}), which also sets the required extent of any fixed parent grid. Note that T_{eff} is the slope of the *actual* source — radial flow blueshifts it well above the freeze-out temperature.

2.3 The logarithmic singularity at q_T^*

Inside the band the window changes topology where η_{lo} reaches zero, i.e. where $C - Q - M$ changes sign (case boundary of (8)). Solving $m_{aT}m_{bT} = C - q_T p_T$ by the same squaring as in Sec. 2.2 — legitimate on the branch $C - q_T p_T \geq 0$, which holds there since $m_{aT}m_{bT} > 0$ — flips only the sign of the linear term in (12), so the roots are $-q_T^{\mp}$, and the positive one is

$$q_T^* = \frac{m_a\big(p_b\, m_{bT} - E_b\, p_T\big)}{m_b^2} = -\,q_T^-, \tag{16}$$

which lies inside the band $(0, q_T^+)$ precisely when $q_T^- < 0$, i.e. $p_T < p_b$.

For the local behavior define $g \equiv Q + M - C$, so $g < 0$ is the two-interval regime ($\eta_{\mathrm{lo}} > 0$) and $g > 0$ the merged one, with $g(q_T^*) = 0$; g is strictly increasing in q_T ($\mathrm{d}g/\mathrm{d}q_T = p_T + q_T\, m_{bT}/m_{aT} > 0$), so near the transition $g \simeq g'(q_T^*)\,(q_T - q_T^*)$. Expand the integrand factors around $\Delta\eta = 0$ using $\cosh\Delta\eta = 1 + \tfrac{1}{2}\Delta\eta^2 + \mathcal{O}(\Delta\eta^4)$:

$$Q + D = Q + M\cosh\Delta\eta - C \;=\; g + \tfrac{1}{2}M\Delta\eta^2 + \mathcal{O}(\Delta\eta^4), \qquad Q - D \;=\; 2Q - g - \tfrac{1}{2}M\Delta\eta^2 \;\simeq\; 2Q, \tag{17}$$

the second factor staying regular (it vanishes at the *other* edge η_{hi}, far from $\Delta\eta = 0$). Hence, up to a cutoff Λ separating the local region from the rest of the window,

$$\int \frac{\mathrm{d}\Delta\eta}{\sqrt{(Q-D)(Q+D)}} \;\simeq\; \frac{1}{\sqrt{2Q}}\int \frac{\mathrm{d}\Delta\eta}{\sqrt{g + \tfrac{1}{2}M\Delta\eta^2}}, \tag{18}$$

and the local integral has elementary antiderivatives on both sides of the transition:

$$g > 0: \quad \int_0^{\Lambda} \frac{\mathrm{d}\Delta\eta}{\sqrt{g + \tfrac{1}{2}M\Delta\eta^2}} = \sqrt{\tfrac{2}{M}}\,\operatorname{arcsinh}\Big(\Lambda\sqrt{\tfrac{M}{2g}}\Big) = \sqrt{\tfrac{2}{M}}\Big[\ln\big(2\Lambda\sqrt{\tfrac{M}{2}}\big) - \tfrac{1}{2}\ln g\Big] + \mathcal{O}(g), \tag{19}$$

$$g < 0: \quad \int_{\eta_{\mathrm{lo}}}^{\Lambda} \frac{\mathrm{d}\Delta\eta}{\sqrt{\tfrac{1}{2}M\big(\Delta\eta^2 - \eta_{\mathrm{lo}}^2\big)}} = \sqrt{\tfrac{2}{M}}\,\operatorname{arccosh}\Big(\frac{\Lambda}{\eta_{\mathrm{lo}}}\Big) = \sqrt{\tfrac{2}{M}}\Big[\ln\big(2\Lambda\sqrt{\tfrac{M}{2}}\big) - \tfrac{1}{2}\ln|g|\Big] + \mathcal{O}(g), \tag{20}$$

where the second line used $g + \frac{1}{2}M\Delta\eta^2 = \frac{1}{2}M(\Delta\eta^2 - \eta_{\rm lo}^2)$ with $\eta_{\rm lo} = \sqrt{-2g/M}$ and $\operatorname{arccosh} x = \ln(2x) + \mathcal{O}(x^{-2})$. Both sides carry the *same* coefficient $-\frac{1}{2}\sqrt{2/M}\,\ln|g|$: the inner integral has a symmetric, integrable logarithmic singularity

$$I(q_T) \;=\; -\frac{1}{\sqrt{2QM}}\,\ln\left|q_T - q_T^*\right| \;+\; (\text{regular}), \tag{21}$$

after inserting the linearization of g. A Gauss rule straddling a logarithm converges only algebraically, so the q_T interval is split at q_T^* whenever $q_T^- < 0 < q_T^* < q_T^+$; the residual $\theta\ln\theta$ endpoint behavior after the cosine map is discussed in Sec. 2.4.

2.4 The cosine substitution and the tensor-product rule

Every one-dimensional interval $[x_{\rm lo}, x_{\rm hi}]$, of midpoint $m = \frac{1}{2}(x_{\rm lo} + x_{\rm hi})$ and half-width $h = \frac{1}{2}(x_{\rm hi} - x_{\rm lo})$, is traversed via

$$x(\theta) = m - h\cos\theta, \qquad \theta \in (0,\pi), \qquad \mathrm{d}x = h\,\sin\theta\,\mathrm{d}\theta, \tag{22}$$

which maps $\theta = 0 \mapsto x_{\rm lo}$ and $\theta = \pi \mapsto x_{\rm hi}$ monotonically. The endpoint expansion is exact in half-angle form:

$$x - x_{\rm lo} = h\,(1 - \cos\theta) = 2h\sin^2\tfrac{\theta}{2}, \qquad x_{\rm hi} - x = h\,(1 + \cos\theta) = 2h\cos^2\tfrac{\theta}{2}. \tag{23}$$

Consider an integrand with the inverse-square-root endpoint behavior established in Sec. 2.1, $f(x) = g(x)/\sqrt{(x - x_{\rm lo})(x_{\rm hi} - x)}$ with g smooth. Using (23) and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$,

$$f\big(x(\theta)\big)\,\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{g\big(x(\theta)\big)\;h\cdot 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\sqrt{2h\sin^2\frac{\theta}{2}\;\cdot\;2h\cos^2\frac{\theta}{2}}} = g\big(x(\theta)\big)\;: \tag{24}$$

the singular factors cancel *identically*, not just to leading order — the θ-integrand is as smooth as g, and Gauss–Legendre in θ converges spectrally. (If only one endpoint is singular the cancellation happens there and the other endpoint merely gains harmless node clustering; endpoint zeros $\propto \sqrt{x - x_{\rm lo}}$, as in the μ integral, become smooth $\propto \sin^2\frac{\theta}{2}$ by the same algebra.)

The θ integral itself is discretized by an n-point Gauss–Legendre rule with nodes $x_i \in (-1,1)$ and weights w_i, mapped affinely to $(0,\pi)$: $\theta_i = \frac{\pi}{2}(x_i + 1)$, $\int_0^\pi G\,\mathrm{d}\theta \approx \frac{\pi}{2}\sum_i w_i G(\theta_i)$. Combining with (22),

$$\int_{x_{\rm lo}}^{x_{\rm hi}} f(x)\,\mathrm{d}x \;\approx\; h\sum_{i=1}^{n} v_i\,f\big(m - h\,c_i\big), \qquad c_i = \cos\theta_i, \quad v_i = \tfrac{\pi}{2}\,w_i\sin\theta_i, \tag{25}$$

so the map costs no trigonometry at evaluation time: the fused pairs (c_i, v_i) are computed once per order and shared (memoized) across all decay objects. Applying (25) in both directions gives the tensor-product kernel sum

$$K(E_b) \;\approx\; \sum_{\rm seg}\; h_q \sum_{i=1}^{n_q} v_i^q\; q_T^{(i)}\; h_\eta\Big(q_T^{(i)}\Big) \sum_{j=1}^{n_\theta} \frac{v_j^\theta}{\sqrt{Q^2 - D_{ij}^2}}\,\big[\text{bracket}\big]_{ij}, \tag{26}$$

where the inner $1/\sqrt{}$ is evaluated in the factored form $(Q-D)(Q+D)$ and the singular endpoint behavior is tamed by (24) through the mapped $\Delta\eta$ nodes. The only residual non-smoothness is the logarithm of Sec. 2.3 at the segment endpoint q_T^*: with $x - q_T^* \propto \theta^2$ from (23), the integrand behaves as $\sin\theta\cdot\ln\theta^2 \sim 2\,\theta\ln\theta$ near $\theta = 0$ — continuous but not analytic — for which the Gauss error decreases algebraically, $\sim n_q^{-4}$ (measured: the error at low p_T drops $\approx 16\times$ per doubling of n_q).

2.5 Three-body: the μ average

For $a \to b + c + d$ the kernel is averaged over the effective mass $\mu \in [m_c + m_d,\, m_a - m_b]$ with weight $p_c^{\tilde{c}}(\mu)$ and normalization $\mathcal{N} = \int \mathrm{d}\mu\, p_b^a(\mu)\, p_c^{\tilde{c}}(\mu)$ [Eqs. (45)–(48)]. The integrand vanishes as a square root at both endpoints: $p_c^{\tilde{c}} \sim \sqrt{\mu - (m_c + m_d)}$ at threshold, and $K \propto$ band width $\propto p_b^a(\mu) \sim \sqrt{m_a - m_b - \mu}$ at the upper limit. The same substitution (22) turns $\sqrt{\ }$-type zeros into smooth θ^2 behavior, and the μ integral is done with an n_μ-point cosine-mapped Gauss rule; the nodes μ_i, energies $E_b(\mu_i)$ and normalized weights are precomputed at construction.

2.6 Measured convergence and default orders

For a Boltzmann source ($T = 0.15\,\mathrm{GeV}$) relative errors against high-order references behave as follows (worst case is low p_T, where the q_T^* logarithm is inside the band):

orders		error at $p_T = 0.2$	error at $p_T = 0.8$
$\rho \to \pi\pi$:	$(n_q, n_\theta) = (128, 32)$	$7 \cdot 10^{-8}$	$5 \cdot 10^{-12}$
	$(64, 16)$	$2 \cdot 10^{-6}$	$5 \cdot 10^{-12}$
	$(32, 8)$	$3 \cdot 10^{-6}$	$2 \cdot 10^{-8}$
$\omega \to 3\pi$:	$(n_\mu, n_q, n_\theta) = (32, 128, 32)$	$2 \cdot 10^{-8}$	$3 \cdot 10^{-13}$
	$(16, 64, 16)$	$5 \cdot 10^{-7}$	$3 \cdot 10^{-10}$
	$(8, 32, 8)$	$1 \cdot 10^{-4}$	$5 \cdot 10^{-5}$

The $\Delta\eta$ direction is spectrally convergent and over-resolved already at $n_\theta = 16$; n_q is limited by the logarithm ($\sim n_q^{-4}$, only relevant for $p_T < p_b$); $n_\mu \geq 16$ is safe while $n_\mu = 4$ fails at the percent level. The defaults $(n_q, n_\theta, n_\mu) = (128, 32, 32)$ are conservative; cost scales linearly in each order. Number conservation, $N_b = B\, N_a$, holds at the 10^{-5} level for both decays at default orders.

3 Extensions: four-body decays and finite widths

The two- and three-body kernels above cover the bulk of a hadronic decay list, but two physical ingredients are missing from the sharp-mass, $\leq$ 3-body treatment: decays with four or more daughters, and the finite widths of the resonances themselves. Both are handled by the *same* device already used for the three-body case — an average of the two-body kernel over an effective-mass distribution — and the key practical question for each is whether the extra integration multiplies the number of (expensive) kernel evaluations or only enriches the (cheap) weight.

3.1 Recursive effective-mass structure

Return to the three-body average and read it structurally. For $a \to b + (cd)$ the observed daughter b recoils against the pair treated as a single effective particle $\widetilde{cd}$ of mass μ; the *angular* distribution of b — hence the entire $(q_T, \Delta\eta)$ kernel — depends on the pair only through μ, via $E_b(\mu) = (m_a^2 + m_b^2 - \mu^2)/2m_a$. The internal breakup of the pair enters solely through *how much probability sits at each* μ, i.e. the invariant-mass density of the pair,

$$\rho_2(\mu; m_c, m_d) \;\propto\; q(\mu; m_c, m_d), \qquad q(M; m_1, m_2) \equiv \frac{\lambda^{1/2}(M^2, m_1^2, m_2^2)}{2M}, \tag{27}$$

which is precisely the weight $p_c^{\tilde{c}}(\mu)$ of Eq. (47). Writing the map this way,

$$\mathcal{F}_b = \frac{B\, m_a}{4\pi\, \mathcal{N}} \int_{\mu_-}^{\mu_+} \mathrm{d}\mu\ \underbrace{p_b^a(\mu)\, \rho_2(\mu)}_{\text{weight}}\ K\big(E_b(\mu)\big), \qquad \mathcal{N} = \int \mathrm{d}\mu\, p_b^a(\mu)\, \rho_2(\mu), \tag{28}$$

makes the generalization immediate. For an n-body decay $a \to b + (c_1 \cdots c_{n-1})$, cluster the $n-1$ unobserved particles into $\widetilde{c}$ of mass $\mu \in [\sum_i m_{c_i},\, m_a - m_b]$; the kernel is unchanged, $K(E_b(\mu))$, and only the weight generalizes to the invariant-mass density $\rho_{n-1}(\mu)$ of the cluster.

Where the density comes from. The density and its recursion are the standard factorization of n-body Lorentz-invariant phase space. Peeling the observed particle b against the cluster of invariant mass μ,

$$\mathrm{d}\Phi_n\big(m_a^2;\, p_b, \{p_{c_i}\}\big) = \frac{\mathrm{d}\mu^2}{2\pi}\ \mathrm{d}\Phi_2\big(m_a^2;\, m_b^2, \mu^2\big)\ \mathrm{d}\Phi_{n-1}\big(\mu^2;\, \{p_{c_i}\}\big), \tag{29}$$

and integrating the two-body factor over the isotropic emission angle, $\int \mathrm{d}\Phi_2(m_a^2; m_b^2, \mu^2) \propto q(m_a; m_b, \mu) = p_b^a(\mu)$, identifies the μ-measure as $p_b^a(\mu)\,\mathrm{d}\mu$ times the cluster's own phase-space volume. Defining that volume as the density, $\rho_{n-1}(\mu)\,\mathrm{d}\mu \equiv \int \mathrm{d}\Phi_{n-1}(\mu^2; \{p_{c_i}\})$ at fixed cluster mass, and applying (29) once more to $\mathrm{d}\Phi_{n-1}$ — peeling c_1 against the sub-cluster of mass σ — gives the Byckling–Kajantie recursion, one particle at a time,

$$\rho_k\big(\mu; m_{c_1}, \ldots, m_{c_k}\big) = \int_{\sigma_-}^{\sigma_+} \mathrm{d}\sigma\ q(\mu; m_{c_1}, \sigma)\ \rho_{k-1}\big(\sigma; m_{c_2}, \ldots, m_{c_k}\big), \qquad \rho_1(\sigma; m) = \delta(\sigma - m), \tag{30}$$

with $\sigma \in [\sum_{i\geq 2} m_{c_i},\, \mu - m_{c_1}]$. The overall constants (2π from (29), the 4π from each angular integration) are μ-independent and cancel in the normalized ratio $\mathcal{F}_b$, so they are dropped. The recursion terminates at $\rho_1 = \delta$ (a single particle has definite mass); one step gives $\rho_2 = q$ (the three-body weight of Eq. 47), and for four-body

$$\rho_3(\mu; m_c, m_d, m_e) = \int_{m_d+m_e}^{\mu - m_c} \mathrm{d}\sigma\ q(\mu; m_c, \sigma)\ q(\sigma; m_d, m_e). \tag{31}$$

The four-body map explicitly. Assembling the pieces, the observed-b spectrum of $a \to b\,c\,d\,e$ is the two-dimensional mass average

$$\mathcal{F}_b = \frac{B\, m_a}{4\pi\, \mathcal{N}} \int_{m_c+m_d+m_e}^{m_a - m_b} \mathrm{d}\mu\ p_b^a(\mu)\ \underbrace{\left[\int_{m_d+m_e}^{\mu - m_c} \mathrm{d}\sigma\ q(\mu; m_c, \sigma)\, q(\sigma; m_d, m_e)\right]}_{\rho_3(\mu)}\ K\big(E_b(\mu)\big), \quad \mathcal{N} = \int \mathrm{d}\mu\, p_b^a(\mu) \rho_3(\mu). \tag{32}$$

The bracket is the Lorentz-invariant phase-space volume of the three-particle cluster $c\,d\,e$ at total mass μ — geometrically, the area of its Dalitz plot. The inner σ integral runs over the (de) sub-mass; by the symmetry of phase space any pairing gives the same ρ_3, so the choice of which particle to peel first is free. Both mass integrals carry the $\sqrt{\ }$-type endpoint zeros of the Källén function (each q vanishes at its threshold), regularized by the cosine map of Sec. 2.4.

Cost. The crucial point is that $\rho_{n-1}(\mu)$ is *independent of the kernel*: it is a static, $(n-2)$-fold nested phase-space integral in the mass variables alone, computed once on the n_μ nodes at construction. The number of expensive kernel evaluations therefore stays n_μ — the *same* as three-body — for any number of unobserved daughters; only the weight becomes a nested (but kernel-free, hence cheap) integral. A four-body decay matrix costs essentially what a three-body one costs. The inner mass integrals inherit the same $\sqrt{\ }$ endpoint zeros as the μ integral (the Källén function vanishes at each threshold and pseudo-threshold), and are handled by the same cosine-mapped Gauss rule of Sec. 2.4. For general n the weight is an $(n-2)$-fold nested mass integral; with n_σ points per level it costs $n_\mu\, n_\sigma^{n-2}$ evaluations of the elementary function q — arithmetic only, no kernel — against the fixed n_μ kernel evaluations. So the build cost is dominated by the kernel until n is large, and a four- or five-body matrix is, in practice, a three-body matrix with a slightly richer weight.

On observing a cluster particle. In the cascade every daughter is observed in turn (Sec. 4). When the observed particle is one of the "cluster" members, one simply relabels: the observed particle plays the role of b and the remaining $n-1$ form the cluster. The construction above is symmetric under that relabelling, so nothing new is required — the observed slot is chosen, the rest are clustered, and ρ_{n-1} is built from their masses.

3.2 Finite widths

The kernels above take every mass sharp. A resonance of width Γ_i instead carries a spectral function $\varrho_i(m)$ — a (relativistic, mass-dependent-width, kinematically truncated and renormalized) Breit–Wigner,

$$\varrho_i(m) \;\propto\; \frac{m^2\,\Gamma_i(m)}{\big(m^2-m_i^2\big)^2+m_i^2\,\Gamma_i(m)^2}, \qquad \int_{m_{\min}}^{m_{\max}} \varrho_i(m)\,\mathrm{d}m = 1, \tag{33}$$

reducing to $\delta(m-m_i)$ as $\Gamma_i \to 0$. For a resonance decaying to two particles in relative orbital angular momentum ℓ the width itself is mass dependent,

$$\Gamma_i(m) = \Gamma_i\,\frac{m_i}{m}\left(\frac{q(m;m_1,m_2)}{q(m_i;m_1,m_2)}\right)^{2\ell+1}\frac{B_\ell\big(q(m)\big)}{B_\ell\big(q(m_i)\big)}, \tag{34}$$

with B_ℓ the Blatt–Weisskopf barrier factor. It vanishes at the threshold $m=m_1+m_2$ (where the daughter momentum $q\to 0$), cutting the low tail of ϱ_i and enforcing that the resonance cannot decay below threshold; the normalization runs over the open range and is recomputed per channel. Finite width enters the decay map in three distinct places, which differ sharply in cost:

1. **Unobserved cluster constituents.** A width on any c_i folds *into the mass density*: in the recursion (30) the sharp $q(\mu;m_{c_1},\sigma)$ is averaged, $q \to \int \mathrm{d}m_{c_1}\,\varrho(m_{c_1})\,q(\mu;m_{c_1},\sigma)$. This is kernel-free — it modifies $\rho_{n-1}(\mu)$ only — and hence *cheap*, exactly like adding a daughter. Physically this is already how an intermediate resonance is represented (e.g. $a\to b\,\rho$, $\rho\to\pi\pi$ *is* the three-body μ integral with ϱ_ρ sharpening the weight near $\mu=m_\rho$).

2. **Observed daughter b.** Here E_b, p_b and the whole q_T band depend on m_b, so a width on b requires an *outer average of the kernel*, $K \to \int \mathrm{d}m_b\,\varrho_b(m_b)\,K$, one full kernel evaluation

per m_b node — *expensive*. In practice the observed daughter is usually stable or narrow (the measured π, K, p, Λ), so this reduces to the sharp δ and costs nothing; it matters only for broad observed resonances mid-cascade.

3. **Parent** a**.** The parent mass sets $C = m_a E_b$, m_{aT}, and the band edges $q_T^{\pm}$; a width on a likewise demands an outer average $\int \mathrm{d}m_a\, \varrho_a(m_a)\,(\cdots)$ with a full kernel per node — *expensive*, but often the most important physically, since the broadest states (σ, ρ, Δ, higher resonances) are parents.

Thus the same organizing principle recurs: widths of particles that enter *only through the effective mass* are absorbed into the precomputed weight at negligible cost, whereas widths of the two particles whose masses enter the *kinematics of the observed leg* (parent and observed daughter) multiply the number of kernel evaluations.

3.2.1 Resonances and widths are one object

The cluster-width folding (case 1) and the effective-mass integral are not two mechanisms but one. Near a resonance R in the $c\,c'$ channel the invariant-mass density $\rho_2(\mu) = q(\mu; m_c, m_{c'})$ is multiplied by the resonance propagator, i.e. it *becomes* a Breit–Wigner peak in μ: the two statements "$a \to b\,R$ with $R \to c\,c'$ of finite width" and "$a \to b\,c\,c'$ with the pair weighted by $\varrho_R(\mu)$" are literally the same μ integral (28). A narrow R sends $\varrho_R \to \delta(\mu - m_R)$ and factorizes the cascade into the sharp two-body steps $a \to bR$ then $R \to cc'$; a broad R smears them together. This is precisely why intermediate-resonance width is the cheap case — it lives entirely in the weight that the effective-mass integral already evaluates — and it is the physically dominant one, since the broad $\rho, \sigma, K^*, \Delta$ appear overwhelmingly as intermediate states.

3.2.2 The parent-width average as a matrix sum

For the matrix representation (Sec. 4) the parent-width case is especially clean. The sharp-mass map is linear and m_a enters only kinematically, so the width average is a weighted sum of sharp-mass matrices,

$$W_n^{\mathrm{eff}} = \int \mathrm{d}m_a\, \varrho_a(m_a)\, W_n(m_a) \;\approx\; \sum_i w_i\, W_n\big(m_a^{(i)}\big), \tag{35}$$

built at n_{m_a} mass-quadrature nodes spanning the truncated support of ϱ_a. The result is a matrix of the *same shape*: the cascade, apply, persistence and streaming machinery are all untouched, and the width is absorbed entirely at build time, once, then persisted. The cost is n_{m_a} sharp builds for that channel. Interpolating W_n in m_a rather than rebuilding is tempting but fails for the same reason a collocation in μ fails: the q_T band $[q_T^-, q_T^+]$ of Eq. (14) *moves* with m_a, so the matrices have shifting support and do not interpolate smoothly — the honest mass-quadrature is required.

Reopening closed channels. A channel with $\sum_{\text{daughters}} m > m_a$ at the nominal parent mass is kinematically *closed* and is skipped by the sharp-mass cascade (Sec. 4). A finite parent width reopens it: on the upper tail $m_a' > \sum m$ of ϱ_a the channel is allowed, and its contribution is $\int_{\sum m}^{m_{\max}} \mathrm{d}m_a'\, \varrho_a(m_a')\,(\cdots)$, weighted by the (small) probability in that tail. This is the physical content of the "closed-at-nominal-mass" count reported by the cascade: those channels carry zero weight in the sharp limit and nonzero weight once the parent spectral function is restored.

Production and decay consistency. One subtlety is worth stating. In a thermal source the parent is produced *with* its spectral function already folded into the density, so the source $\mathcal{F}_a$ carries ϱ_a in its normalization. The decay then requires the parent's mass distribution *conditional* on its momentum. The standard approximation takes the production and decay spectral functions identical — the same ϱ_a in both places — which is exact in the narrow-width limit and a controlled approximation otherwise. A fully consistent treatment integrates the parent four-momentum and its mass jointly; the factorized form used here, $\mathcal{F}_a$ times the ϱ_a-averaged decay (35), is the usual on-shell-spectral compromise, adequate while Γ_a/m_a is not too large.

Status. Four-body is now implemented (the 531 unique cluster keys of the list's 400 four-body channels): the constructor reuses the entire three-body kernel/matrix path and only swaps the pair density $p_c^{\tilde{c}}$ for the nested weight $\rho_3(\mu)$ of (31) — an inner cosine-mapped σ-quadrature — exactly as argued here. It is validated by four-body number conservation and by the cluster-mass permutation symmetry of ρ_3 (agreement at machine precision, $\sim 10^{-16}$). Finite width remains future work: it needs an outer spectral average for the parent/observed legs (35) (or an enriched weight for the rest). Finite width is not a small correction for this list: of the 1518 species, 713 carry a width and 464 are broad ($\Gamma > 150\,\mathrm{MeV}$), and many broad states are parents — so the expensive parent-average applies to hundreds of channels, and persistence (Sec. 4) becomes essential rather than optional. The cost accounting sets the implementation priority: cluster/intermediate widths first (cheap, and — being the broad $\rho, \sigma, K^*, \Delta$ — the dominant feed-down effect), parent width second (expensive but important for the broad primaries), observed-daughter width last (rare).

4 Boost-invariant sources: one matrix per mode

Let the parent mode $\mathcal{F}_{a,n}$ be independent of η_q and sampled on a grid $q_T^{(1)}, \ldots, q_T^{(N_q)}$, and let interpolation reconstruct it anywhere via local Lagrange cardinal weights L_j of order P (stencil of P points, error $\mathcal{O}(h^P)$),

$$\mathcal{F}_{a,n}(q_T) \;\approx\; \sum_j L_j(q_T)\, \mathcal{F}_{a,n}\big(q_T^{(j)}\big), \qquad L_j\big(q_T^{(i)}\big) = \delta_{ij}. \tag{36}$$

First the mode bookkeeping. Insert the mode input $\mathcal{F}_a(q_T, \eta_q, \phi_q) = \mathcal{F}_{a,n}(q_T)\cos(n\phi_q)$ into the source bracket of (1), evaluated at $\phi_p = 0$; together with the two rapidity signs (equal contributions, since neither the kernel nor this source distinguishes $\pm\Delta\eta$) the four source evaluations are

$$\sum_{s=\pm}\sum_{r=\pm} \mathcal{F}_{a,n}(q_T)\cos\big(n(0{+}r\alpha)\big) = \mathcal{F}_{a,n}(q_T)\left[2\cos(n\alpha){+}2\cos(-n\alpha)\right] = 4\cos(n\alpha)\,\mathcal{F}_{a,n}(q_T) = 4\,T_n(Z)\,\mathcal{F}_{a,n}(q_T), \tag{37}$$

using the evenness of the cosine and $\cos(n \arccos Z) = T_n(Z)$ [Eq. (40)]. Now discretize (1) by the tensor rule (26) and insert the interpolation (36) at every q_T node:

$$\mathcal{F}_{b,n}\big(p_T^{(k)}\big) \approx \frac{B\, m_a}{4\pi p_b} \sum_{\text{seg}} h_q \sum_i v_i^q\, q_T^{(i)}\, h_\eta^{(i)} \sum_{j'} \frac{v_{j'}^\theta}{\sqrt{Q^2 - D_{ij'}^2}}\, 4T_n(Z_{ij'}) \underbrace{\sum_j L_j\big(q_T^{(i)}\big)\, \mathcal{F}_{a,n}\big(q_T^{(j)}\big)}_{\text{interpolated source}}$$

$$= \sum_{j=1}^{N_q} \underbrace{\left[\frac{B\, m_a}{4\pi p_b} \sum_{\text{seg}} h_q \sum_i v_i^q\, q_T^{(i)}\, h_\eta^{(i)} \sum_{j'} \frac{v_{j'}^\theta\, 4T_n(Z_{ij'})}{\sqrt{Q^2 - D_{ij'}^2}}\, L_j\big(q_T^{(i)}\big) \right]}_{= W_n[k,j]} \mathcal{F}_{a,n}\big(q_T^{(j)}\big), \tag{38}$$

where the second line only exchanged the finite sums — legitimate because every factor except $\mathcal{F}_{a,n}$ is independent of the source. The matrix element $W_n[k,j]$ is thus the exact quadrature of the kernel against the j-th cardinal function. Assembly therefore runs the *same* quadrature loops as a functor evaluation; wherever the source would be evaluated, the bracketed multiplier is instead scattered onto the P columns of the interpolation stencil of $q_T^{(i)}$ ("scatter assembly"). One row costs one kernel evaluation with no source calls; the whole matrix costs N_{p_T} of them. Equivalently — and this was used as a cross-check — column j of W_n equals the functor applied to the j-th cardinal basis function, by the same linearity; the scatter form merely avoids the N_q-fold redundancy of computing columns one at a time.

Two properties of the quadrature matter for the matrix:

- The Θ-support in q_T is the band $q_T^\pm = m_a\big(E_{b|c}^a\, p_T \pm p_{b|c}^a\, m_{bT}\big)/m_b^2$; each row is supported (up to the stencil halo) inside its band. For light daughters the band is wide ($\propto m_b^{-2}$), so in practice the matrices are effectively dense (measured fill 65–76% for $\rho \to \pi\pi$, $\omega \to \pi\pi\pi$), and they are numerically full rank; neither sparse nor low-rank storage pays. At $N_{p_T} \times N_q \sim 39 \times 121$ a matrix is $\sim 37\,\text{KB}$ and applies in microseconds.

- The integrand has an integrable $\ln|q_T - q_T^*|$ singularity at $q_T^* = -q_T^-$, where the $\Delta\eta$ domain splits; the interval is divided there. All $1/\sqrt{\ }$ endpoint singularities (in $\Delta\eta$, at $Z = \pm 1$) and $\sqrt{\ }$-type endpoint zeros (in μ) are absorbed by the substitution $x = \text{mid} - \text{half}\cos\theta$, whose $\sin\theta$ Jacobian regularizes them; fixed-order Gauss–Legendre then converges spectrally except for the residual $\theta \ln\theta$ behavior at q_T^* ($\sim n_q^{-4}$).

The only error introduced relative to the functor is interpolation, $\mathcal{O}(h^P)$; with a cubic stencil ($P = 4$) on $h = 0.035\,\text{GeV}$ a thermal source reproduces the functor to a few $\times 10^{-5}$, and the truncation "$\mathcal{F}_a \equiv 0$ outside the grid" requires the source to be negligible beyond the last grid point.

4.1 The Z-space form of the assembly integral

For the matrix assembly the inner rapidity integral can be freed of all transcendental functions. The observation is that an η-independent source never requires the rapidity $\Delta\eta$ itself — only $Z = D/Q$ enters, through the Chebyshev factor $T_n(Z)$.

4.1.1 Derivation

Start from the inner integral of the mode-n assembly at fixed q_T, with both rapidity signs folded in (factor 2, legitimate because the kernel is even in $\Delta\eta$ and the source does not depend on it) and both azimuthal roots summed (the $2T_n$ of Eq. 41):

$$I_n \;=\; 2\int_{\eta_{\rm lo}}^{\eta_{\rm hi}} \frac{\mathrm{d}\Delta\eta}{\sqrt{Q^2-D^2}}\, 2\,T_n(Z), \qquad D = M\cosh\Delta\eta - C, \quad Z = \frac{D}{Q}. \tag{39}$$

Substitute $c = \cosh\Delta\eta$, $\mathrm{d}\Delta\eta = \mathrm{d}c/\sqrt{c^2-1}$, which maps the window (8) to $c \in [\,\max(1, (C-Q)/M),\, (C+Q)/M\,]$; then pass to

$$Z = \frac{Mc - C}{Q}, \qquad c(Z) = \frac{C + QZ}{M}, \qquad \mathrm{d}c = \frac{Q}{M}\,\mathrm{d}Z, \qquad \sqrt{Q^2-D^2} = Q\sqrt{1-Z^2}. \tag{40}$$

The window endpoints become $Z = 1$ (kinematic edge, $c = (C+Q)/M$) and $Z = Z_{\rm lo}$ with

$$Z_{\rm lo} = \max\!\left(-1,\; \frac{M-C}{Q}\right), \tag{41}$$

the value $(M-C)/Q$ corresponding to $c = 1$, i.e. $\Delta\eta = 0$. Collecting the Jacobians,

$$I_n = \frac{4}{M}\int_{Z_{\rm lo}}^{1} \frac{T_n(Z)\,\mathrm{d}Z}{\sqrt{(1-Z^2)\,(c(Z)^2-1)}}. \tag{42}$$

This is an exact change of variables, not an approximation: the $\Delta\eta$ form of Sec. 2 and (42) are the same number. What has changed is the computational content of one quadrature node: $c(Z)$ is a multiply–add, $T_n(Z)$ follows from the recurrence $T_{k+1} = 2ZT_k - T_{k-1}$ [Eq. (40)], and the only remaining special function is one square root. No arccosh (window bounds), cosh (per node) or arccos (azimuthal angle) appears.

4.1.2 Quadrature: two regimes

Full window, $Z_{\rm lo} = -1$ (i.e. $C - Q \geq M$). The factor $1/\sqrt{1-Z^2}$ is exactly the Chebyshev–Gauss weight, so with the standard nodes the weight is integrated *exactly*:

$$I_n \;\approx\; \frac{4}{M}\,\frac{\pi}{n_\theta}\sum_{k=1}^{n_\theta} \frac{T_n(Z_k)}{\sqrt{c(Z_k)^2-1}}, \qquad Z_k = \cos\theta_k, \quad \theta_k = \frac{(2k-1)\pi}{2n_\theta}. \tag{43}$$

The nodes are precomputed once per order; note also $T_n(Z_k) = \cos(n\theta_k)$, so for fixed n even the Chebyshev factors are constants of the rule.

Partial window, $-1 < Z_{\rm lo} < 1$ (i.e. $C - Q < M$). Both endpoint factors vanish like square roots *inside* the domain: $1 - Z^2$ at $Z = 1$, and $c^2 - 1$ at $Z = Z_{\rm lo}$. The decisive point is that $c - 1 = \frac{Q}{M}(Z - Z_{\rm lo})$ is *exactly linear* in Z (immediate from $c(Z) = (C+QZ)/M$ and $C + QZ_{\rm lo} = M$), so the full radicand splits into an endpoint-singular part and a manifestly regular remainder,

$$\big(1-Z^2\big)\big(c^2-1\big) = \underbrace{(1-Z)(Z-Z_{\rm lo})}_{\text{endpoint zeros}}\;\underbrace{\tfrac{Q}{M}\,(1+Z)\,(c+1)}_{\equiv\; R(Z)^2}, \tag{44}$$

with $R(Z)^2$ *strictly positive on the closed interval* ($Q, M > 0$; $1 + Z > 0$ because $Z \geq Z_{\rm lo} > -1$; $c + 1 \geq 2$). The integrand is therefore exactly the form of Sec. 2.4, $f(Z) = g(Z)/\sqrt{(Z - Z_{\rm lo})(1 - Z)}$ with the *smooth* numerator

$$g(Z) = \frac{T_n(Z)}{R(Z)} = \frac{T_n(Z)}{\sqrt{\frac{Q}{M}(1 + Z)(c + 1)}}. \tag{45}$$

So g carries the *shifted Chebyshev weight* $1/\sqrt{(Z - Z_{\rm lo})(1 - Z)}$ on $[Z_{\rm lo}, 1]$ — the identical structure to the full window (43), only on a shifted interval — and is integrated by the *same* rule: shifted Gauss–Chebyshev, whose nodes carry the weight exactly and whose weights are the uniform π/n_θ,

$$I_n \;\approx\; \frac{4}{M}\,\frac{\pi}{n_\theta}\sum_{k=1}^{n_\theta}\frac{T_n(Z_k)}{R(Z_k)}, \qquad Z_k = m + h\cos\theta_k, \quad \theta_k = \frac{(2k-1)\pi}{2n_\theta}, \quad c_k = \frac{C + QZ_k}{M}, \tag{46}$$

m, h the midpoint and half-width of $[Z_{\rm lo}, 1]$. Equations (43) and (46) are *one formula* $I_n = \frac{4}{M}\frac{\pi}{n_\theta}\sum_k T_n(Z_k)/R(Z_k)$ on the two windows, differing only in the interval ($[-1, 1]$ vs $[Z_{\rm lo}, 1]$) and in the regular factor left in R: $R^2 = c^2 - 1$ for the full window, $R^2 = \frac{Q}{M}(1+Z)(c+1)$ for the partial. Both remove the inverse-square-root weight *analytically*, so the summand is bounded, no positivity guard is needed, and the near-edge nodes carry their correct finite contribution.

Transition zone. As $Z_{\rm lo} \to -1^+$ the smooth g acquires a square-root branch point at $Z = -1$ (from the $1+Z$ in R) just below the lower endpoint, and the fixed-order Gauss–Chebyshev error degrades — at $Z_{\rm lo} = -0.995$, $n_\theta = 32$ gives relative error $\sim 10^{-3}$. Quadrupling the node count restores it ($\sim 10^{-12}$): the partial branch densifies for $Z_{\rm lo} < -\frac{1}{2}$, exactly as the full branch does (Eq. 43 region) on the other side of the boundary.

The algebraically equivalent alternative — forming $v_i\,T_n(Z_i)/\sqrt{(1 - Z_i^2)(c_i^2 - 1)}$ with the fused weight $v_i = \frac{\pi}{2}w_i \sin\theta_i$ of (25) carrying the $\sin\theta$ against a radicand that itself vanishes at the edge — is the same number in exact arithmetic (indeed $h\,v_i/\sqrt{(1 - Z^2)(c^2 - 1)} = \frac{\pi}{2}w_i/R(Z)$), but in floating point it is a 0/0: the radicand rounds to $\lesssim 0$ for roughly 1% of nodes (worst case $\mathcal{O}(10^{-31})$ negative), where the square root would fault or return ∞. It must then be guarded ($u > 0$ or skip), which silently forfeits those nodes. The factored form (46) needs neither the guard nor the sacrifice — $R(Z)$ is regular by construction.

4.1.3 Convergence and the transition zone

The integrand of (43), $f(Z) = T_n(Z)/\sqrt{c^2 - 1}$, has a branch point at $Z = Z_{\rm lo}$, *outside* the interval when $Z_{\rm lo} < -1$. For a function analytic on $[-1, 1]$ with nearest singularity at a real point $Z_0 < -1$, Gauss–Chebyshev error decays geometrically at the rate of the Bernstein ellipse through Z_0,

$$\varepsilon \;\sim\; \rho^{-2n_\theta}, \qquad \rho = |Z_0| + \sqrt{Z_0^2 - 1}. \tag{47}$$

Evaluating (47) with $Z_0 = Z_{\rm lo}$ and the working order $n_\theta = 8$:

$Z_{\rm lo}$	-3	-1.5	-1.2	-1.05
ρ	5.83	2.62	1.86	1.37
ρ^{-2n_θ}	$6 \cdot 10^{-13}$	$2 \cdot 10^{-7}$	$4.7 \cdot 10^{-5}$	$6.5 \cdot 10^{-3}$

The rule is excellent while the branch point stays away from the interval and degrades sharply as $Z_{\rm lo} \to -1^+$: the "transition zone". This failure mode was first seen empirically: the $n = 2$ matrix validation degraded to 2.3×10^{-3} while $n = 0$ remained at 10^{-4}. The asymmetry is instructive — for $n = 0$ the under-resolution errors are of a single sign in Z and largely average out under the subsequent q_T integration, whereas the sign-changing T_2 weight prevents the cancellation and exposes them. Per (47), quadrupling the order multiplies the exponent by four (ρ^{-8n_θ}), e.g. at $Z_{\rm lo} = -1.05$: $1.37^{-64} \approx 2 \cdot 10^{-9}$. The implementation therefore switches to $4n_\theta$ Chebyshev nodes whenever $Z_{\rm lo} > -\frac{3}{2}$, i.e. exactly where (47) predicts the base rule falls above the 10^{-7} level. The choice of threshold is a cost/accuracy compromise: a first attempt with the densification band extended to $Z_{\rm lo} > -3$ was accurate but slowed the cascade below the gain of the Z-space form itself.

4.1.4 Measured effect

On the full 1518-species cascade (8 threads, production orders n_q=32, n_θ=8, n_μ=8, 168-point grid):

assembly scheme	single 2-body build	cascade	n=2 validation
$\Delta\eta$-space (arccosh + cosh + arccos)	0.51 ms	20.2 s	$2.6 \cdot 10^{-5}$
$\Delta\eta$-space, arccos $\to T_n$ recurrence	0.41 ms	20.2 s	$2.6 \cdot 10^{-5}$
Z-space, naive	0.27 ms	16.8 s	$2.3 \cdot 10^{-3}$ (fails)
Z-space, densified transition	0.33 ms	18.3 s	$1.3 \cdot 10^{-4}$

This form applies only to the assembly (and any η-independent evaluation): the functor path evaluates an arbitrary source at $\eta_p \pm \Delta\eta$ and $\phi_p \pm \alpha$, so there the rapidity and the azimuthal angle are genuine arguments and the $\Delta\eta$-space form of Sec. 2 remains the appropriate one.

5 Rapidity-dependent sources: the $\Delta\eta$-Toeplitz structure

5.1 The kernel as a rapidity convolution

The boost-invariant result of Sec. 4 folded the $\pm\Delta\eta$ contributions together because the source carried no rapidity dependence. For a general source that fold is not available; instead the rapidity structure of the kernel must be exposed. Return to the mode-n decay map *before* any η-folding. The analytic azimuthal integration [Eq. (41)] gives, for a rapidity-dependent source,

$$\mathcal{F}_{b,n}(p_T, \eta_p) = \frac{B\, m_a}{4\pi p_b} \int_0^\infty \mathrm{d}q_T\, q_T \int_{-\infty}^\infty \mathrm{d}\Delta\eta\, \frac{\Theta\big(Q^2 - D^2\big)}{\sqrt{Q^2 - D^2}}\, 2T_n(Z)\, \mathcal{F}_{a,n}\big(q_T, \eta_p + \Delta\eta\big), \tag{48}$$

where the parent rapidity has been written $\eta_q = \eta_p + \Delta\eta$ and $\mathrm{d}\eta_q = \mathrm{d}\Delta\eta$ at fixed η_p; every kinematic quantity $D = M\cosh\Delta\eta - C$, $Z = D/Q$ depends on rapidity only through $\Delta\eta$. The daughter rapidity η_p therefore appears in (48) *only* as the shift of the source's argument — the statement that the decay map commutes with longitudinal boosts, $\eta_p \to \eta_p + \text{const}$ carrying η_q along with it.

Collect everything that multiplies the source into a *profile*

$$K_n(p_T, q_T, \Delta\eta) \;\equiv\; \frac{B\, m_a}{4\pi p_b}\, q_T\, \frac{\Theta\big(Q^2 - D^2\big)}{\sqrt{Q^2 - D^2}}\, 2T_n(Z(q_T, \Delta\eta)), \tag{49}$$

so that (48) reads

$$\mathcal{F}_{b,n}(p_T,\eta_p)=\int_0^\infty \mathrm{d}q_T\int_{-\infty}^{\infty}\mathrm{d}\Delta\eta\; K_n(p_T,q_T,\Delta\eta)\;\mathcal{F}_{a,n}(q_T,\eta_p+\Delta\eta)\,: \tag{50}$$

at each (p_T,q_T) this is a *convolution* of the source in rapidity with the fixed profile $K_n(p_T,q_T,\cdot)$. Three properties of the profile follow directly from the kernel analysis of Sec. 2.1 and carry the entire structure of what follows:

1. **Even:** $K_n(\,\cdot,-\Delta\eta)=K_n(\,\cdot,\Delta\eta)$, since D, Z, and $\Theta(Q^2-D^2)$ all depend on $\Delta\eta$ through $\cosh\Delta\eta$ alone. The profile is a symmetric rapidity point-spread function; it does *not* know the sign of the boost.

2. **Compact support:** K_n vanishes outside $|\Delta\eta|\in[\eta_{\rm lo},\eta_{\rm hi}]$ (the window of Sec. 2.1), a set of total rapidity extent $\le 2\,\eta_{\rm hi}=2\,\mathrm{arccosh}[(C+Q)/M]$, in practice 1–2 units.

3. **η_p-independent:** K_n depends on p_T and q_T but not on where in rapidity the daughter sits. This is boost invariance, and it is what makes (50) a genuine convolution rather than a general integral operator.

Physically $K_n(p_T,q_T,\cdot)$ is the rapidity smearing induced by the decay: a parent line at a single rapidity feeding daughter momentum p_T is spread by the profile into a band of daughter rapidities of width set by $\eta_{\rm hi}$. A boost-invariant source ($\mathcal{F}_{a,n}$ independent of η_q) is the eigenfunction with the source pulling out of the convolution, $\mathcal{F}_{b,n}=\big(\int K_n\,\mathrm{d}\Delta\eta\big)\mathcal{F}_{a,n}$, recovering Sec. 4 — there the $\int K_n\,\mathrm{d}\Delta\eta$ *is* the boost-invariant matrix element.

5.2 Discretization: the block-Toeplitz matrix

Sample both spectra on a common *uniform* rapidity grid $\eta_a=\eta_0+a\,h_\eta$ (uniformity is essential — it is what turns the convolution into a Toeplitz operator). Index the daughter spectrum by $(k,a)\leftrightarrow(p_T^{(k)},\eta_a)$ and the parent by $(j,b)\leftrightarrow(q_T^{(j)},\eta_b)$. Discretize (50) in two stages. First the q_T integral and profile, by the same quadrature-plus-interpolation as Sec. 4, which produces the q_T scatter onto column index j. Second the source's rapidity argument: at a quadrature node with offset $\Delta\eta$, the value $\mathcal{F}_{a,n}(q_T^{(j)},\eta_a+\Delta\eta)$ is reconstructed from the η grid by the uniform Lagrange stencil,

$$\mathcal{F}_{a,n}\big(q_T^{(j)},\eta_a+\Delta\eta\big)\;\approx\;\sum_b L^\eta\big(\eta_a+\Delta\eta;\,\eta_b\big)\,\mathcal{F}_{a,n}\big(q_T^{(j)},\eta_b\big). \tag{51}$$

The crucial observation is that on a uniform grid the cardinal weight depends only on the *offset* of the evaluation point from the node in grid units,

$$L^\eta\big(\eta_a+\Delta\eta;\,\eta_b\big)=\ell\Big(\frac{(\eta_a+\Delta\eta)-\eta_b}{h_\eta}\Big)=\ell\Big(\frac{\Delta\eta}{h_\eta}-(b-a)\Big), \tag{52}$$

a function of $b-a$ and $\Delta\eta$ but *not* of a and b separately — translation invariance of the uniform grid, inherited from translation invariance of the profile. Assembling the coefficient of $\mathcal{F}_{a,n}(q_T^{(j)},\eta_b)$ in $\mathcal{F}_{b,n}(p_T^{(k)},\eta_a)$ therefore yields a quantity that depends on the rapidity indices only through $d\equiv b-a$:

$$W\big[(k,a),(j,b)\big]=w\big[k,\,j,\,d\big],\qquad d=b-a, \tag{53}$$

with

$$w[k,j,d] = \int \mathrm{d}\Delta\eta\; \widetilde{K}_n^{(k,j)}(\Delta\eta)\, \ell\Big(\tfrac{\Delta\eta}{h_\eta} - d\Big), \tag{54}$$

$\widetilde{K}_n^{(k,j)}$ being the profile after the q_T scatter (the quadrature sum over the q_T nodes whose stencil hits column j). Viewing W as an $N_\eta \times N_\eta$ array of $N_{p_T} \times N_q$ blocks $B_d = w[\cdot,\cdot,d]$, (53) says W is constant along block diagonals,

$$W = \begin{pmatrix} B_0 & B_1 & B_2 & \cdots \\ B_{-1} & B_0 & B_1 & \\ B_{-2} & B_{-1} & B_0 & \\ \vdots & & & \ddots \end{pmatrix}, \tag{55}$$

a *block-Toeplitz* matrix. Storing W densely repeats each block up to N_η times; storing the tensor $w[k,j,d]$ once is exact and lossless — it is the discrete profile, the point-spread function of Sec. 5.1 sampled at integer rapidity offsets.

Two further exact properties shrink the tensor, both inherited from the profile:

1. **Banded:** compact support of K_n (item 2 above) plus the finite reach of the η stencil limits the nonzero offsets to $|d| \le L$ with $L = \lceil \eta_{\mathrm{hi}}^{\max}/h_\eta \rceil + \frac{P}{2}$ (window half-width plus stencil halo), $\eta_{\mathrm{hi}}^{\max} = \max_{q_T} \operatorname{arccosh}[(C+Q)/M]$, in practice 1–2 units; so $B_d = 0$ for $|d| > L$ and only $2L+1$ offsets exist.

2. **Even in d:** evenness of the profile (item 1) gives $w[k,j,-d] = w[k,j,d]$, so only $d \ge 0$ need be stored. (The *source* need not be symmetric in rapidity; the *profile* is.)

For $N_{p_T} = 39$, $N_q = 121$, $N_\eta = 61$, $h_\eta = 0.05$, $L \approx 40$: dense W is $(39\cdot61)\times(121\cdot61) \approx 1.4\times10^8$ entries ($\sim$140 MB), the tensor is $39 \times 121 \times 41 \approx 1.9 \times 10^5$ entries ($\sim$1.5 MB) — a factor $\sim 10^2$, at zero accuracy cost, because it encodes exact symmetries (translation invariance and parity of the profile) rather than a numerical truncation.

Two further exact properties shrink the tensor:

1. **Banded:** the Θ-support limits $|\Delta\eta| \le \operatorname{arccosh}\big[(C+Q)/M\big]$, in practice 1–2 units of rapidity, so $B_d = 0$ for $|d| > L$ with $L = \lceil \Delta\eta_{\max}/h_\eta \rceil$; only $2L+1$ offsets exist.

2. **Even in d:** the kernel weight depends on $\Delta\eta$ only through $\cosh\Delta\eta$, so $w[k,j,d] = w[k,j,-d]$ and only $d \ge 0$ need be stored. (The *source* need not be symmetric; the *profile* is.)

For $N_{p_T} = 39$, $N_q = 121$, $N_\eta = 61$, $h_\eta = 0.05$, $L \approx 40$: dense W is $(39\cdot61)\times(121\cdot61) \approx 1.4\times10^8$ entries ($\sim$140 MB), the tensor is $39\times121\times41 \approx 1.9\times10^5$ entries ($\sim$1.5 MB) — a factor $\sim 10^2$, at zero accuracy cost, because it encodes an exact symmetry rather than a numerical truncation.

5.3 Assembly

The assembly generalizes the scatter of Sec. 4 by one stencil dimension. Fix a Fourier mode n and run the same $(q_T, \Delta\eta)$ quadrature loops as the kernel (in $\Delta\eta$ space — see the remark below); a node at $(q_T, \Delta\eta)$ with total quadrature factor

$$\mathrm{fac} = \mathrm{pref}\cdot v_i^q h_q\, q_T\, h_\eta\, \frac{v_j^\theta}{\sqrt{Q^2 - D^2}}\, 2T_n(Z) \tag{56}$$

(the 2 is the two azimuthal roots only — the $\pm\Delta\eta$ fold is *not* performed here, each sign now lands on different η columns) deposits into the tensor via the product of the q_T stencil and the η-offset stencil:

$$w[k,\, j_0+m-1,\, d_0+m'-1] \mathrel{+}= \text{fac}\; L^q_m(q_T)\; L^\eta_{m'}\big(\Delta\eta/h_\eta\big), \qquad m,m'=1,\ldots,P, \tag{57}$$

where d_0 is the stencil start of the offset $\Delta\eta/h_\eta$ on the integer offset axis. Since the kernel factor is even in $\Delta\eta$, only $\Delta\eta \geq 0$ need be traversed and the result stored for $d \geq 0$; the negative offsets are recovered at application time. No loop over η_p exists anywhere: one sweep per (k,j) column of nodes builds the profile valid for *every* daughter rapidity, so the assembly cost equals that of the boost-invariant matrix times the P-fold extra stencil work — *not* times N_η. This is the payoff of the Toeplitz structure on the assembly side.

Two remarks. First, the Z-space form of Sec. 4.1 does *not* apply here: the η-offset stencil needs the rapidity $\Delta\eta$ itself, so the assembly stays in $\Delta\eta$ space (or would have to pay an arccosh per node to recover $\Delta\eta$ from c — exactly the function the Z form eliminates). Second, the offset variable is interpolated on the *uniform* η grid, so the uniform-grid Lagrange weights apply verbatim; the interpolation error in the η direction is the same $\mathcal{O}(h_\eta^P)$ as in q_T.

5.4 Application

Using the evenness $w[k,j,-d] = w[k,j,d]$, the apply is a folded discrete convolution over the stored $d \geq 0$ half:

$$\mathcal{F}_b\big(p_T^{(k)},\eta_a\big) = \sum_j \Big[\, w[k,j,0]\, \mathcal{F}_a\big(q_T^{(j)},\eta_a\big) + \sum_{d=1}^{L} w[k,j,d]\, \big(\mathcal{F}_a\big(q_T^{(j)},\eta_{a+d}\big) + \mathcal{F}_a\big(q_T^{(j)},\eta_{a-d}\big)\big)\Big]. \tag{58}$$

Organizing the computation per offset makes each term a dense $(N_{p_T} \times N_q)$ matrix acting on an η-shifted copy of the source — $L+1$ small GEMMs over the $(q_T \times \eta)$ source array:

$$F_b \;=\; \sum_{d=0}^{L} B_d\, \big[S^{+d}F_a + (1-\delta_{d0})\, S^{-d}F_a\big], \qquad B_d = w[\,\cdot,\cdot,d\,], \tag{59}$$

with $S^{\pm d}$ the shift along η. Cost: $\mathcal{O}(N_{p_T}N_qN_\eta L)$ multiply–adds ($\approx 2\times10^7 \sim$ milliseconds at N_{p_T}=39, N_q=121, N_η=61, $L\approx$40), organized as BLAS-3 rather than the BLAS-2 of a dense block matrix. For large N_η the Toeplitz structure admits FFT application along η, $\mathcal{O}(N_{p_T}N_qN_\eta \log N_\eta)$ — worthwhile only once $L \gg \log N_\eta$, i.e. for very fine rapidity grids; at the sizes above the direct fold wins.

For azimuthally asymmetric, rapidity-dependent sources the procedure is per mode as before: one tensor w_n per Fourier mode, assembled with $2T_n(Z)$, applied to the mode functions $\mathcal{F}_{a,n}(q_T,\eta_q)$.

5.5 Edges and grid layout

Translation invariance fails only at the ends of the rapidity grid: daughter points within $\Delta\eta_{\max}$ of the boundary require parent values beyond it. Since physical sources vanish rapidly at large $|\eta|$ (e.g. $e^{-m_T\cosh\eta/T}$), the consistent convention is zero-padding — the same "source vanishes off the grid" convention used in q_T — with the η grid extended by $\Delta\eta_{\max} = \max_{q_T} \operatorname{arccosh}[(C+Q)/M]$ (evaluated at the band edge; typically 1–2 units) beyond the rapidity window where results are

needed. Concretely: to output $|\eta_p| \le \eta^{\mathrm{obs}}$, the parent grid must span $|\eta_q| \lesssim \eta^{\mathrm{obs}} + \Delta\eta_{\max}$, and the truncation error inherits the source's decay at the padding edge — for thermal-like sources this is $e^{-m_{aT}(\cosh(\eta^{\mathrm{obs}}+\Delta\eta_{\max})-1)/T}$, negligible already for modest padding.

The memory layout follows the apply: storing w as (k, j)-major slices per offset d (i.e. $L{+}1$ dense $N_{p_T} \times N_q$ matrices B_d) makes each term of (58) a contiguous GEMM; `Float32` storage is appropriate for the same reason as in Sec. 4 (rounding 6×10^{-8}, far below the quadrature setting), halving the $\sim$ 1–3 MB per decay and mode.

6 Summary

case	object	size / cost
η-independent, per mode n	dense $W_n \in \mathbb{R}^{N_{p_T} \times N_q}$	$\sim$40 KB, apply $\sim \mu$s
η-dependent, per mode n	tensor $w[k, j, d]$, $\lvert d\rvert \le L$, $d \ge 0$	$\sim$1 MB, apply $\sim$ms

Decay chains compose by matrix multiplication (and, in the Toeplitz case, by convolution of profiles), so the total feed-down of a hadron list is itself a precomputed linear map.